# Connor Tynan

+44 7724 870844 | connor.tynan@outlook.com | LinkedIn | GitHub | Website | ORCID

## Education

**University of Exeter** Sept 2019 – June 2023
Master of Mathematics (MMath) Degree (Hons) – First Class (75.6 %) Exeter, UK

## Skills

| | |
|---|---|
| Programming: | Python, MATLAB, R |
| Libraries & Frameworks: | pandas, NumPy, scikit-learn, seaborn, matplotlib, PyTorch |
| Tools: | AWS, git, HPC, Jupyter, Docker |

## Experience

**Higher Scientist (Data Science)** June 2025 – Present
National Physical Laboratory (NPL) London, UK

- Developed software dedicated to implementation of statistical methods, e.g., synthetic data generation using neural networks, etc.
- Investigated correlated stochastic processes and the effects of such on uncertainty estimates.
- Adopted role of technical lead on customer-facing projects.

**Scientist (Data Science)** Mar 2024 – June 2025
National Physical Laboratory (NPL) London, UK

- Awarded Early Career Scientist Award 2024 for Best Presentation, titled “”
- Developed software for generating synthetic tabular data, implementing statistical methods, and performing numerical analysis.
- Contributed to numerous works, including reports on inverse problems, hypothesis testing and trustworthy AI.
- Reviewed, updated, and maintained existing uncertainty quantification software.
- Performed technical reviews of peers’ work.
- Investigated the effects of correlated stochastic processes in improving uncertainty estimates.

## Projects

**Mathematically Modelling Microbial Interactions** – GitHub Sept 2022 – May 2023
University of Exeter Exeter, UK

- Performed an extensive literature review, researched existing models and reproduced results to establish an initial understanding of the field
- Preprocessed and contextualised novel microbial growth data before fitting to an existing mathematical model using the Hamiltonian Monte Carlo algorithm, validating fits using MCMC convergence diagnostics
- Investigated underlying dynamics using data analysis, steady-state analysis and bifurcation theory, finding a transcritical bifurcation to underpin changes in interactions

**Predicting Student Success** – GitHub Oct 2023 – Nov 2023

- Established an understanding of relevant business application before performing exploratory data analysis to better comprehend the data; namely, univariate and multivariate analysis of categorical and numerical features
- Evaluated the performance of 9 multiclass classification models using several evaluation metrics; the best performing model was chosen and underwent hyperparameter tuning for optimal performance
- Served the final model using Flask and waitress, containerised and isolated the project using pipenv and Docker before, finally, deploying the Docker image to a cloud service using AWS Elastic Beanstalk

**SMS Spam Detection Using NLP** – GitHub | Kaggle Nov 2023 – Dec 2023

- Explored and analysed raw text data using pandas and matplotlib before preprocessing the data using NLP techniques, such as tokenization, lemmatization and vectorization
- Compared the performance of several supervised learning algorithms and tuned the best performing model using a grid search approach, improving the accuracy and AUC scores slightly
- Configured an AWS Lambda function to utilise the final model and deployed a REST API using AWS API Gateway to connect to and test the Lambda function